

Invariant subvarieties with small dynamical degree

(Joint works with Matsuzawa, Meng, Shibata & Zhang)

Guolei Zhong

Center for Complex Geometry, Institute for Basic Science

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Background & Motivation

- We work over an algebraically closed field of characteristic zero.
- Let $f : X \dashrightarrow X$ be a dominant rational map of a projective variety.

One of the most fundamental questions in the dynamical system is to study the distribution of points and subvarieties.

- **periodic points**: Can we bound the number or get an asymptotic upper bound after iteration?
- **non-periodic points**: Can we understand the distribution by measuring the density of forward orbits?

Definition (The first dynamical degree)

The **first dynamical degree** is defined by

$$d_1(f) := \lim_{n \rightarrow \infty} ((f^n)^* H \cdot H^{\dim(X)-1})^{1/n}$$

where H is a nef and big Cartier divisor of X .

Background & Motivation

Some remarks on the dynamical degree

- (Dinh-Sibony, 2005) showed that $d_1(f)$ is a birational invariant and is independent of the choice of H .
- For a dominant rational map of a (not necessarily projective) variety, by taking any birational projective model, $d_1(f)$ is well-defined.
- If f is an endomorphism of a projective variety, $d_1(f)$ coincides with the spectral radius of $f^*|_{N^1(X)}$.

Example

- Let $f : X \rightarrow X$ be a polarized endomorphism of a projective variety, i.e., $f^*H \sim qH$ for some ample divisor H . Then $d_1(f) = q$.
- (Cantat, 1999) Let $f : S \rightarrow S$ be an automorphism of a compact Kähler surface. If $d_1(f) > 1$, then S is one of (1) non-minimal rational surface; (2) K3 surface; (3) Enriques surface; or (4) complex torus.

Background & Motivation

Conjecture (Shou-Wu Zhang, 2006)

Let f be q -polarized of a projective variety X with $q > 1$. Then

$$\lim_{n \rightarrow \infty} \# \text{Fix}(f^n) = q^{n \cdot \dim(X)} (1 + o(1))$$

Remarks on Zhang's conjecture

- (Fakruddin, 2003) The set of periodic points are countable and Zariski dense.
- (Zhang, 2006) ✓ if X is smooth and general fixed points have multiplicity 1.
- (Dinh-Nguyễn-Trung, 2016) ✓ if X is smooth.
- Unknown for singular case, no complete result even in $\dim(X) = 2$.
- Two important examples: projective spaces with the coordinate map; abelian varieties with the multiplication map.

Background & Motivation

- In general, there are infinitely many f -invariant subvarieties of positive dimension. For example, let $X := \mathbb{P}^1 \times \mathbb{P}^1$ and $f := f_1 \times \text{id}$.
- Inspired from arithmetic geometry, it is natural to study the finiteness of invariant subvarieties of small dynamical degree (SDD).

SDD subvarieties (Matsuzawa-Meng-Shibata-Zhang-Z., 2022)

Let $f : X \rightarrow X$ be a dominant self-morphism of an algebraic variety.

- An (irreducible) closed subvariety $V \subseteq X$ is **f -periodic** (of period s) if $\overline{f^s(V)} = V$ for some $s > 0$.
- An f -periodic closed subvariety V (of period s) is of **small dynamical degree** if $d_1(f^s|_V) < d_1(f^s)$. Σ_{f^∞} : set of f -periodic SDD subvarieties
- $V \in \Sigma_{f^\infty}$ is **maximal** if for any $V' \in \Sigma_{f^\infty}$ with $V \subseteq V'$, we have $V = V'$. $S_{f^\infty} \subseteq \Sigma_{f^\infty}$: subset of maximal elements
- $P_f \subseteq S_f \subseteq S_{f^\infty}$: subset of f -invariant elements and prime divisors

Example: when f is q -polarized with $q > 1$, S_f consists of points.

Main results

Question (MMSZZ, 2022)

Let $f : X \rightarrow X$ be a dominant self-morphism of an algebraic variety X .

- (1) Is S_f a finite set? Namely, are there only finitely many f -invariant maximal f -periodic subvarieties of small dynamical degree?
- (2) Is P_f a finite set? Namely, are there only finitely many prime divisors in S_f ?
- (3) Does the upper bound (of cardinality)

$$\#S_{f^n} \leq (d_1(f))^{n \cdot \dim(X)}(1 + o(1))$$

hold as $n \rightarrow \infty$?

- Question (3) is an extension of Shou-Wu Zhang's conjecture.
- In ([Matsuzawa-Meng-Shibata-Zhang, 2023](#)), Question (1) is applied to study the equivalence of Kawaguchi-Silverman conjecture and small Arithmetic Non-Density conjecture.

Main results

Theorem A (MMSZZ, 2022)

Let f be a dominant self-morphism of a variety X . Then

$$\lim_{n \rightarrow \infty} \#S_{f^n} \leq (d_1(f))^{n \cdot \dim(X)}(1 + o(1)),$$

if (X, f) fits one of the following cases.

- (1) X is an algebraic group; and $f = R_a \circ g$ where g is an isogeny and R_a is the right multiplication map by some $a \in X$ (✓ X is semi-abelian).
- (2) X is a toric variety with the big torus T being f -invariant.

Theorem B (MMSZZ, 2022)

Let f be a surjective endomorphism of a projective variety. Then

$$\lim_{n \rightarrow \infty} \#P_{f^n} \leq (d_1(f))^n(1 + o(1)).$$

Main results

Theorem C (MMSZZ, 2022)

S_f is finite for any surjective endomorphism f of a projective surface or a Q -abelian variety (quasi-étale quotient of an abelian variety).

If the nef cone of a projective variety X is “nice”, then we found the following (in progress):

- There is an equivariant morphism $\pi : X \rightarrow Y$ such that any surjective endomorphism f descends to a polarized endomorphism g on Y .
- There is an “almost” bijective map $S_{f^n} \rightarrow \text{Fix}(g^n)$ so that we reduce the asymptotic upper bound on $\#S_{f^n}$ to counting the periodic points.

Theorem D (Meng-Z., 2023)

If $f : X \rightarrow X$ is a surjective endomorphism of a Mori dream space with $d_1(f) > 1$, then S_f is finite.

- The only thing left is to extend (Dinh-Nguyễn-Trung, 2016) to the mildly singular variety (in progress).

Thank you for your attention!